Literature Review: Patterns in U.S. Federal Contract Awards

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1 Introduction and Project Context

This project analyzes federal contract award records from USAspending.gov, distributed in a rolling seven-day snapshot through the Open FedSpend repository [2]. My saved data set contains 27,729 records and variables that describe recipients, awarding agencies, obligations, industries, states where work is performed, and contract dates. The data is a short-period snapshot rather than a full historical record. Award amounts in this data refer to obligations, not necessarily total contract ceilings.

The project asks which recipients and industries obtain large contract awards, which agencies issue the most awards, how amounts differ across agencies, which states account for the greatest award values, and whether contract duration is associated with award amount.

2 Research Literature

2.1 Omari et al. (2025): Historical federal procurement data

Omari, Alansari, Libgober, and Kaufman present a cleaned collection of nearly 100 million federal procurement contract actions spanning 1979–2023. They describe the difficulty of obtaining and interpreting procurement records and provide data resources for studying federal contracting patterns [4].

This study relates to my research because it examines federal contracting patterns using many of the same variables, including awarding agencies, contract values, and recipients. However, while the researchers analyzed decades of procurement data, my project focuses on a much smaller snapshot

of federal awards. Their research demonstrates how access to a larger historical dataset can reveal long-term trends that my analysis cannot fully capture.

2.2 Parker and Roy (2024): Industries and recipient characteristics

Parker and Roy analyze 1,551,610 federal contracts across 58 agencies and compare contract awards by STEM intensity, product or service category, and business ownership characteristics. Their findings associate higher-STEM-intensity contracts with lower representation of women-owned and minority-owned businesses [6].

This study connects to my research question about which recipients and industries receive federal contracts. Both projects examine how federal awards differ across industries, but Parker and Roy focus specifically on disparities involving women-owned and minority-owned businesses. My analysis takes a broader approach by examining recipients, industry classifications, and award amounts without focusing on business ownership demographics.

2.3 Blackford (2026): Geographic distribution of federal awards

Blackford reviews federal contract spending across states, focusing on small businesses and trends over fiscal years 1997–2023. The Congressional Research Service report finds persistent geographic patterns, with large-state economies and locations linked to major federal or defense activity often receiving substantial contracting dollars [1].

This report is particularly relevant to my research question about which states receive the greatest value in federal contract awards. Blackford examines geographic patterns over several years, while my project looks at the distribution of awards within a much shorter period. Comparing these approaches highlights how geographic spending patterns may vary depending on the time frame and whether the analysis focuses on all contractors or specifically small businesses.

2.4 Fazekas, Ferrali, and Wachs (2023): Agency differences and favoritism risk

Fazekas, Ferrali, and Wachs study federal contracts and campaign donations from 2004–2015. Using statistical designs and constructed favoritism-risk indicators, they report that larger political donations are associated with increased risks of preferential treatment, especially in more politically influenced agencies [3].

This research relates to my questions about how federal contract awards and spending differ between government agencies. Although both studies examine agency-level contracting activity, researchers investigate how political contributions can influence contracting decisions. My project instead focuses on identifying patterns in the number and value of awards across agencies, without attempting to determine the political factors behind those decisions.

3 Synthesis and Relevance to My Project

*O*verall, these studies provide a better understanding of how federal contracting patterns can be analyzed across agencies, industries, recipients, and geographic locations. Although each study approaches federal procurement from a different perspective, they demonstrate the value of using contract data to identify spending patterns and differences in award distribution. My project builds on these ideas by examining similar variables within a smaller dataset of federal contract awards. Unlike the previous studies, my analysis also explores the relationship between contract duration and award amounts. While the limited time frame of my dataset makes it difficult to identify long-term trends, it provides an opportunity to explore current contracting patterns and identify areas for further research.

4 AI Use Disclosure

ChatGPT was used to help verify bibliographic information and explain the LaTeX submission process [5]. The comparison of the literature with my own research questions is written in my own words.

References

[1] R. Corinne Blackford. Geography of small business contracts: Where federal procurement dollars are awarded to small businesses. Technical Report R48900, Congressional Research Service, March 2026. Report text: https://www.everycrsreport.com/reports/R48900.html.

[2] ConorsCode. Open fedspend data: U.s. federal contract awards, rolling seven-day window. GitHub dataset repository, originating from USAspending.gov, 2026. Local snapshot analyzed: 27,729 records; July 29–August 4, 2026. Dataset is updated over time. Source: https://github.com/ConorsCode/open-fedspend-data.

[3] Mihály Fazekas, Romain Ferrali, and Johannes Wachs. Agency independence, campaign contributions, and favoritism in US federal government contracting. *Journal of Public Administration Research and Theory*, 33(2):262–278, 2023. DOI: https://doi.org/10.1093/jopart/muac026.

[4] Aymane Omari, Nasser Alansari, Brian Libgober, and Aaron R. Kaufman. A comprehensive dataset of united states federal procurement from 1979 to 2023. *Scientific Data*, 12:1368, 2025. DOI: https://doi.org/10.1038/s41597-025-05714-1.

[5] OpenAI. Chatgpt. AI language model, 2026. Prompt: "Can you please help walk me through this assignment so I can make sure it is up to the submission standards?" Used to find relevant research reports, verify citation details, and explain LaTeX submission requirements.

[6] Chris Parker and Dwaipayan Roy. A granular examination of gender and racial disparities in federal procurement. *Data & Policy*, 6:e25, 2024. DOI: https://doi.org/10.1017/dap.2024.17.